

# Production Dashboard API — API Specification Design (API仕様設計)

003\_DD-API — supports 003\_DD; handler designed in 003\_DD-FN; requirements REQ-028–REQ-042.

## Document control (改版履歴)

| Field             | Value                  |
|-------------------|------------------------|
| Document ID       | 003_DD-API             |
| System name       | ProductionManagementAI |
| Subsystem name    | Production orders      |
| Work item         | WI-004                 |
| Created by        | Claude (for ThanhTN)   |
| Created date      | 2026-09-22             |
| Last updated by   | Claude (for ThanhTN)   |
| Last updated date | 2026-09-22             |

| Version | Date       | Author               | Revision content                                                                                                                                                                                                              |
|---------|------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | 2026-09-22 | Claude (for ThanhTN) | Initial creation                                                                                                                                                                                                              |
| 2       | 2026-09-22 | Claude (for ThanhTN) | <code>GET /api/system/health</code> added (DEC-017, DEC-019, DEC-020); <code>workload[].weekStart</code> clarified for the current week                                                                                       |
| 3       | 2026-09-23 | Claude (for ThanhTN) | WI-005: the UI is Japanese. Example product names are Japanese automobile parts (WI-005 DEC-002); the one quoted UI text gives the Japanese with an English gloss (WI-005 DEC-003). No field, rule, API or processing changes |

## Overview and operation catalog

| Field             | Value                                                                                                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API / module name | Production dashboard API                                                                                                                                                            |
| Overview          | One read-only JSON endpoint behind SCR-003 that returns every widget's figures from one consistent snapshot (DEC-011, DEC-015). It takes no parameters and writes nothing (REQ-039) |

| No | Endpoint / method      | Handler                                                     | Purpose                                        | Notes                   |
|----|------------------------|-------------------------------------------------------------|------------------------------------------------|-------------------------|
| 1  | GET /api/dashboard     | DashboardController.Get → DashboardService.GetSnapshotAsync | The dashboard snapshot                         | New in WI-004           |
| 2  | GET /api/system/health | SystemController.Health → SystemHealthService.CheckAsync    | Database reachability for the health indicator | New in WI-004 (DEC-017) |

Common rules, unchanged from 001\_DD-API and 002\_DD-API:

- **Auth:** cookie session (0002\_ADR) plus policy `ProductionOrderEditor` = role `Admin` or `Operator`. No session → 401 with an empty body; session without the role → 403 with an empty body. The existing policy is reused for the same reason 002\_DD-API gives: a separate read-only policy would imply a permission split that WI-001 DEC-015 has not decided.
- **Content type:** `application/json`; errors `application/problem+json` (RFC 9457) with the 001\_DD-API shape. No stack trace or exception text is ever returned.
- **JSON:** camelCase; dates `YYYY-MM-DD` (plant-local); timestamps ISO 8601 UTC with `Z`; enums as strings.
- **CSRF:** not applicable — `GET`, no state change, no body (WI-002 DEC-020 posture unchanged).
- **Caching:** the response carries `Cache-Control: no-store`. The figures are a point-in-time snapshot, so a browser or proxy must never answer a later visit to `/` from an earlier snapshot.

## System-wide API registry entries

| ID         | Endpoint               | Overview                                       | Notes                                                                                          |
|------------|------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------|
| API-DSH-01 | GET /api/dashboard     | Production dashboard snapshot                  | New; first entry of the dashboard group                                                        |
| API-SYS-01 | GET /api/system/health | Server and database status for signed-in users | New; the anonymous liveness <code>/health</code> is not an API-registry entry and is unchanged |

`API-PRD-01` and `API-PO-01` – `API-PO-04` stay as registered in 001\_DD-API and 002\_DD-API.

## Shared response object: DashboardOrder

One element of `overdue.orders` or `dueSoon.orders`. A deliberately narrow row: the dashboard shows these fields and links nowhere (DEC-006), so it gets neither `isOverdue` (the group says so), `updatedAt`, notes nor a version.

| No | Name         | Variable name            | Type          | Length | Required | Repeats (array) | Value mapping                 | Example                    | Description                                                                                            | Notes                                                  |
|----|--------------|--------------------------|---------------|--------|----------|-----------------|-------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 1  | ID           | <code>id</code>          | string (uuid) | 36     | yes      | no              | —                             | <code>0197e4a0-...</code>  | <code>production_orders.id</code>                                                                      | Used only as a rendering key; never displayed          |
| 2  | Order number | <code>orderNumber</code> | string        | 13     | yes      | no              | —                             | <code>P0-2026-00022</code> |                                                                                                        |                                                        |
| 3  | Product      | <code>product</code>     | object        | —      | yes      | no              | Join to <code>products</code> | —                          | <code>{ "id", "sku", "name" }</code> — the same <code>ProductSummary</code> as <code>002_DD-API</code> | Rendered "{sku} — {name}" ( <code>003_BD M-12</code> ) |
| 4  | Quantity     | <code>quantity</code>    | integer       | —      | yes      | no              | —                             | <code>2390</code>          |                                                                                                        |                                                        |
| 5  | Due date     | <code>dueDate</code>     | string (date) | 10     | yes      | no              | —                             | <code>2026-09-12</code>    | Plant-local                                                                                            | <code>003_BD M-15</code>                               |
| 6  | Status       | <code>status</code>      | string        | —      | yes      | no              | Enum name                     | <code>InProgress</code>    | Always <code>Draft</code> or <code>InProgress</code> here                                              | Label client-side ( <code>M-11</code> )                |

## Shared response object: OrderGroup

| No | Name   | Variable name       | Type                          | Required | Repeats (array) | Description                                                                         |
|----|--------|---------------------|-------------------------------|----------|-----------------|-------------------------------------------------------------------------------------|
| 1  | Total  | <code>total</code>  | integer                       | yes      | no              | Every order in the group, not only those returned                                   |
| 2  | Orders | <code>orders</code> | <code>DashboardOrder[]</code> | yes      | yes             | At most 10, by due date then order number ( <code>D-01</code> , <code>D-02</code> ) |

## Shared response object: CountAndQuantity

| No | Name        | Variable name           | Type            | Required | Description                                                                                                                                                                |
|----|-------------|-------------------------|-----------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Order count | <code>orderCount</code> | integer         | yes      | 0 when none                                                                                                                                                                |
| 2  | Quantity    | <code>quantity</code>   | integer (int64) | yes      | Sum of <code>quantity</code> ; 0 when none. 64-bit: a sum of 32-bit quantities can exceed $2^{31}$ , and stays exactly representable in a JavaScript number up to $2^{53}$ |

# Endpoint / method design

## 1. GET /api/dashboard

| Field                   | Value                                                                                                                                               |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Description             | Return every figure on SCR-003 computed from one consistent snapshot, with the snapshot time and the plant-local date the figures were computed for |
| Return type             | DashboardResponse                                                                                                                                   |
| Created by / date       | Claude / 2026-09-22                                                                                                                                 |
| Last modified by / date | —                                                                                                                                                   |

**Arguments:** none. The endpoint binds no route, query or body parameter; a query string sent with the request is ignored and cannot change any window or size (003\_BD §5). There is therefore no 400 response.

Processing overview: authorize, then delegate to `DashboardService.GetSnapshotAsync` (003\_DD-FN §1). It reads the clock once, derives every window from the plant-local date (003\_DD-FN §2), runs 003\_DB's seven statements in one read-only `REPEATABLE READ` transaction (§3) and shapes the result (§4). The response is complete or it is an error — never partial (REQ-028).

### Processing flow

| Step | Description                                                                      | Calls                                                         |
|------|----------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1    | Authorize                                                                        | policy <code>ProductionOrderEditor</code>                     |
| 2    | Build the snapshot                                                               | <code>DashboardService.GetSnapshotAsync</code> → 003_DD-FN §1 |
| 3    | 200 with <code>DashboardResponse</code> and <code>Cache-Control: no-store</code> | —                                                             |

### Return value

| Type                           | Name | Description         |
|--------------------------------|------|---------------------|
| <code>DashboardResponse</code> | body | See Response fields |

**Request fields:** not applicable — the endpoint takes no input.

**Response fields** — `DashboardResponse`

| No  | Name            | Variable name                | Type                    | Length | Required | Repeats (array)  | Value mapping                      | Example                           | Description                                                                                              | Notes                                                                                          |
|-----|-----------------|------------------------------|-------------------------|--------|----------|------------------|------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1   | Snapshot time   | <code>asOf</code>            | string (date-time)      | —      | yes      | no               | UTC                                | <code>2026-09-22T05:05:12Z</code> | The instant the clock was read                                                                           | Shown in plant time (M-17)                                                                     |
| 2   | Plant today     | <code>today</code>           | string (date)           | 10     | yes      | no               | T                                  | <code>2026-09-22</code>           | The date every figure was computed for                                                                   |                                                                                                |
| 3   | Plant time zone | <code>timeZone</code>        | string                  | —      | yes      | no               | <code>PlantOptions.TimeZone</code> | <code>Asia/Tokyo</code>           | Lets the client render <code>asOf</code> in the same zone the figures use, rather than the browser's     |                                                                                                |
| 4   | Status counts   | <code>statusCounts</code>    | object                  | —      | yes      | no               | Q1                                 | —                                 | <code>{ total, draft, inProgress, completed, cancelled } ,</code> integers, a status with no orders is 0 | REQ-029; <code>total</code> = sum of the four                                                  |
| 5   | Overdue         | <code>overdue</code>         | <code>OrderGroup</code> | —      | yes      | no               | Q3a                                | —                                 | D-01                                                                                                     | REQ-030                                                                                        |
| 6   | Due soon        | <code>dueSoon</code>         | <code>OrderGroup</code> | —      | yes      | no               | Q3b                                | —                                 | D-02                                                                                                     | REQ-030                                                                                        |
| 7   | Workload        | <code>workload</code>        | object[]                | —      | yes      | yes (exactly 10) | Q2, zero-filled                    | —                                 | D-03 buckets in display order                                                                            | REQ-031, DEC-009                                                                               |
| 7.1 | Bucket kind     | <code>workload[].kind</code> | string                  | —      | yes      | —                | —                                  | <code>week</code>                 | <code>overdue   week   later</code>                                                                      | First is <code>overdue</code> , then $8 \times$ <code>week</code> , last is <code>later</code> |

| No  | Name                   | Variable name                                               | Type                                              | Length | Required | Repeats (array) | Value mapping | Example                            | Description                                                                                                                                                                                         | Notes                                       |
|-----|------------------------|-------------------------------------------------------------|---------------------------------------------------|--------|----------|-----------------|---------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 7.2 | Week start             | <code>workload[].weekStart</code>                           | string (date)   null                              | 10     | yes      | —               | —             | <code>2026-09-28</code>            | First date the bucket covers: the Monday of the week, except the current week, whose first date is <code>today</code> (its earlier days are overdue, D-03); <code>null</code> for overdue and later |                                             |
| 7.3 | Week end               | <code>workload[].weekEnd</code>                             | string (date)   null                              | 10     | yes      | —               | —             | <code>2026-10-04</code>            | Sunday of the week; <code>null</code> for overdue and later                                                                                                                                         | The table equivalent shows the range (M-16) |
| 7.4 | Order count / quantity | <code>workload[].orderCount</code> , <code>.quantity</code> | integer, integer (int64)                          | —      | yes      | —               | —             | <code>7</code> , <code>3150</code> |                                                                                                                                                                                                     | <code>CountAndQuantity</code> fields        |
| 8   | Top products           | <code>topProducts</code>                                    | object[]                                          | —      | yes      | yes (0–10)      | Q4            | —                                  | D-04, open quantity descending, then SKU                                                                                                                                                            | REQ-032                                     |
| 8.1 | Product                | <code>topProducts[].product</code>                          | <code>ProductSummary</code>                       | —      | yes      | —               | —             | —                                  |                                                                                                                                                                                                     |                                             |
| 8.2 | Open quantity          | <code>topProducts[].openQuantity</code>                     | integer (int64)                                   | —      | yes      | —               | —             | <code>2390</code>                  |                                                                                                                                                                                                     |                                             |
| 8.3 | Active orders          | <code>topProducts[].activeOrderCount</code>                 | integer                                           | —      | yes      | —               | —             | <code>2</code>                     |                                                                                                                                                                                                     |                                             |
| 9   | Completed this week    | <code>completedThisWeek</code>                              | <code>CountAndQuantity</code> + <code>from</code> | —      | yes      | no              | Q5            | —                                  | D-05; <code>from</code> = W(T)                                                                                                                                                                      | REQ-034                                     |

| No | Name                 | Variable name      | Type                    | Length | Required | Repeats (array)  | Value mapping   | Example | Description                                                                                                                                                                     | Notes   |
|----|----------------------|--------------------|-------------------------|--------|----------|------------------|-----------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 10 | Completed this month | completedThisMonth | CountAndQuantity + from | —      | yes      | no               | Q5              | —       | D-06; from = 1st of T's month                                                                                                                                                   | REQ-034 |
| 11 | On time              | onTime             | object                  | —      | yes      | no               | Q5              | —       | { onTimeCount, completedCount, windowStart };<br>windowStart = T – 29. The rate is not sent: the client derives it (M-13), so there is one rounding rule and one place it lives | REQ-035 |
| 12 | Lead time            | leadTime           | object                  | —      | yes      | no               | Q5              | —       | { averageDays, orderCount, windowStart };<br>averageDays is the mean rounded half away from zero to one decimal by the server, or null when orderCount = 0                      | REQ-037 |
| 13 | Completion trend     | completionTrend    | object[]                | —      | yes      | yes (exactly 12) | Q6, zero-filled | —       | { weekStart, weekEnd, orderCount }, oldest first; the last is the current week                                                                                                  | REQ-036 |

The `from` and `windowStart` dates are sent so the table equivalents and the tile captions can state their windows exactly, without the client recomputing the plant calendar.

Example response (abridged):

```

{
  "asOf": "2026-09-22T05:05:12Z",
  "today": "2026-09-22",
  "timeZone": "Asia/Tokyo",
  "statusCounts": { "total": 124, "draft": 35, "InProgress": 25, "completed": 56, "cancelled": 8 },
  "overdue": {
    "total": 15,
    "orders": [
      {
        "id": "0197e4a0-0000-7000-8001-000000000078",
        "orderNumber": "PO-2026-00078",
        "product": { "id": "0197e4a0-0000-7000-8000-000000000107", "sku": "P-1007", "name": "ハブベアリング" },
        "quantity": 470,
        "dueDate": "2026-09-08",
        "status": "InProgress"
      }
    ]
  },
  "dueSoon": { "total": 8, "orders": [] },
  "workload": [
    { "kind": "overdue", "weekStart": null, "weekEnd": null, "orderCount": 15, "quantity": 9612 },
    { "kind": "week", "weekStart": "2026-09-22", "weekEnd": "2026-09-27", "orderCount": 6, "quantity": 3270 },
    { "kind": "later", "weekStart": null, "weekEnd": null, "orderCount": 2, "quantity": 2250 }
  ],
  "topProducts": [
    { "product": { "id": "0197e4a0-0000-7000-8000-0000000001029", "sku": "P-1029", "name": "ドアヒンジ" }, "openQuantity": 6750, "activeOrderCount": 4 }
  ],
  "completedThisWeek": { "orderCount": 3, "quantity": 1024, "from": "2026-09-21" },
  "completedThisMonth": { "orderCount": 24, "quantity": 19410, "from": "2026-09-01" },
  "onTime": { "onTimeCount": 23, "completedCount": 30, "windowStart": "2026-08-24" },

```

```

"leadTime": { "averageDays": 13.7, "orderCount": 30, "windowStart": "2026-08-24" },
"completionTrend": [
  { "weekStart": "2026-06-29", "weekEnd": "2026-07-05", "orderCount": 3 },
  { "weekStart": "2026-09-21", "weekEnd": "2026-09-27", "orderCount": 3 }
]
}

```

(`workload` always has 10 entries and `completionTrend` 12; the example shows the first and last of each. Figures are illustrative, not the seed's exact values.)

### Error codes

| Code     | Meaning                                | HTTP status |
|----------|----------------------------------------|-------------|
| —        | Not signed in                          | 401         |
| —        | Missing role (client shows MSG-E021)   | 403         |
| MSG-E013 | Unexpected failure (database, timeout) | 500         |

No 400: there is no input to reject. No 404, 409 or 422: the endpoint reads a snapshot of whatever exists, including nothing (REQ-028's empty state is a 200 with zeros).

## 2. GET /api/system/health

| Field                   | Value                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------------|
| Description             | Report whether the database answers a trivial round trip, for SCR-003's health indicator (003_BD HS-01–HS-05) |
| Return type             | <code>SystemHealthResponse</code>                                                                             |
| Created by / date       | Claude / 2026-09-22                                                                                           |
| Last modified by / date | —                                                                                                             |

**Arguments:** none; a query string is ignored.

Processing overview: authorize with the same policy as the dashboard, run `SELECT 1` with a 2-second timeout (003\_DD-FN §6), and answer **200 in both outcomes**. That the endpoint answered at all is the server's status; the body carries only the database's. Requests to this path never renew the sign-in cookie (DEC-019, 003\_DD-FN §7).

## Processing flow

| Step | Description                                                                                                      | Calls                                                      |
|------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 1    | Authorize                                                                                                        | policy <code>ProductionOrderEditor</code>                  |
| 2    | Ping the database                                                                                                | <code>SystemHealthService.CheckAsync</code> → 003_DD-FN §6 |
| 3    | 200 with <code>SystemHealthResponse</code> and <code>Cache-Control: no-store</code> ; the cookie is not reissued | —                                                          |

**Request fields:** not applicable.

## Response fields — `SystemHealthResponse`

| No | Name       | Variable name          | Type               | Length | Required | Repeats (array) | Value mapping                              | Example                           | Description                                                                          | Notes                                              |
|----|------------|------------------------|--------------------|--------|----------|-----------------|--------------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------|
| 1  | Database   | <code>database</code>  | string             | —      | yes      | no              | <code>ok</code>   <code>unavailable</code> | <code>ok</code>                   | <code>unavailable</code> when the ping failed for any reason or took longer than 2 s | The only component the server reports on (DEC-020) |
| 2  | Checked at | <code>checkedAt</code> | string (date-time) | —      | yes      | no              | UTC                                        | <code>2026-09-22T05:05:30Z</code> | When the server ran the check                                                        | Shown in plant time (003_BD M-20)                  |

Nothing else is returned: no version, host, connection string, latency or exception text, in either outcome.

```
{ "database": "unavailable", "checkedAt": "2026-09-22T05:05:30Z" }
```

## Error codes

| Code                  | Meaning                                                                           | HTTP status                                               |
|-----------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------|
| —                     | Not signed in (including a session that expired because polling did not renew it) | 401                                                       |
| —                     | Missing role                                                                      | 403                                                       |
| <code>MSG-E013</code> | Unexpected failure outside the ping itself                                        | 500 — the browser shows 「サーバー：接続不可」 (Server: Unreachable) |

A failing ping is a 200 with `database: "unavailable"` , not an error: the server answered, and reporting that is the endpoint's job.

## Unresolved decisions

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None. The contract rests on WI-004 DEC-006 (read-only), DEC-008/DEC-009 (windows, buckets), DEC-011 (one endpoint) and DEC-015 (snapshot), and on WI-002 DEC-011/DEC-017 (plant clock), DEC-020 (CSRF) and DEC-023 (RFC 9457). The reuse of the `ProductionOrderEditor` policy is the limitation 002\_DD-API already records against WI-001 DEC-015.